# Operating Systems · 21CSC202J

Unit 1 · Session 11 · SLO 1 — Solved Worksheet

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## Unit â€“ 1 : Session â€“ 11 : SLO â€“ 1 | Steps Involved in Executing a System Call â€” Gap Fill

### I. Completed Sentences (System Call Execution Steps)

1. The user program pushes the system call PARAMETERS onto the stack or into registers.
2. The program invokes the LIBRARY wrapper function for the system call.
3. The wrapper places the system call NUMBER in a register.
4. A TRAP (software interrupt) instruction switches the CPU from user mode to KERNEL MODE.
5. The kernel uses the SYSTEM CALL TABLE to find the address of the service routine.
6. The service routine executes with KERNEL privileges.
7. On completion, the RETURN VALUE is placed in a register.
8. The CPU switches back to USER MODE and the library wrapper returns to the caller.
9. The user program CONTINUES execution after the system call.